

MUHAMMAD RAFEY TAHIR

📍 L'Aquila, Italy 📞 +39-3284415046 ✉ rafeyt21@gmail.com 🌐 [RafeyTahir](#)

EDUCATION

Erasmus Mundus Joint Master Degree on Electric Vehicle Propulsion & Control

Università degli Studi dell'Aquila, Italy.

Feb 2026 - Present

École Centrale de Nantes, France.

Sep 2025 - Jan 2026

Core Courses: Fundamentals of Electrical Vehicle System, Electrical Vehicle Modelling and Simulation, Hybrid Systems Control and Simulation & Nonlinear Systems.

National University of Sciences & Technology, Pakistan.

Aug 2021 - Jan 2024

Masters in Electrical Engineering (Control Systems) **GPA: 3.70**

Core Courses: Power Electronics, Optimal Control, Convex Optimization, Nonlinear Control Systems & Adaptive Control.

Muhammad Nawaz Sharif University of Engineering & Technology, Pakistan.

Oct 2017 - July 2021

Bachelor of Science in Electrical Engineering, **GPA: 3.61** (4th Position)

- Award-winning Final Year Project.
- Ranked in the top 10% of Engineering Science students.
- Registered Engineer **PEC : ELECT/94875**

Core Courses: Linear Control Systems, Digital Logic Design, Microprocessor Systems, Integrated Electronics, Digital Signal Processing.

PUBLICATIONS

M. R. Tahir, M. A. Murtaza, R. Hayat, S. Hutchinson and U. Ali, "A Hierarchical Model-Free Controller for Stable Limit Cycle Generation in Underactuated Mechanical Systems," in *American Control Conference (ACC)*, 2026. [[IEEE Xplore](#)]

WORK & RESEARCH EXPERIENCE

Research Officer - Al-Khawarizmi Institute of Computer Science (KICS)

Dec 2024 – Aug 2025

- Contributed to the development of a 6-DOF robotic manipulator (**Dexter 6 Pro**), developing forward and inverse kinematics solvers and trajectory optimization algorithms for path planning, improving end-effector precision and motion efficiency in dynamic task environments.
- Developed impedance and admittance control schemes for the manipulator, enabling compliant interaction with the environment and safe force-regulated operation in contact-rich tasks.
- Implemented EtherCAT communication protocols for real-time, high-bandwidth data exchange between the controller and joint drives, enabling reliable operation of high-DOF manipulators in industrial settings.

Product Excellence Officer - SkyElectric Pvt. Ltd

March 2024 – Dec 2024

- Collaborated with the Software and Cloud teams to develop an AI-based monitoring system for solar inverters and battery energy storage systems, targeting performance optimization and fault resilience through analysis of electrical parameters including voltage, current, frequency, and harmonics.
- Conducted systematic performance evaluations of inverter systems under varying load and grid conditions, covering power quality analysis, fault detection, and grid stability assessment.

Research Assistant – Control Systems Laboratory - SEECS, NUST

Sep 2022 – Feb 2024

Advisor: [Dr. Usman Ali](#)

- Designed and implemented a novel model-free robust adaptive control framework for underactuated mechanical systems, validated in hardware experiments on a Quanser QNET Rotary Inverted Pendulum Board 2.0 and a Vertical Take-off and Landing (VTOL) system 2.0.
- Conducted research on controller design and optimization techniques for underactuated systems, with a focus on stability analysis and real-time performance under model uncertainty.
- Investigated circuit design automation using reinforcement learning as an exploratory research direction within the lab.
- Assembled a 6-DOF robot manipulator from mechanical components, gaining hands-on experience in system integration and hardware commissioning.

TEACHING EXPERIENCE

National University of Sciences & Technology

- Teaching Assistant, MATH-816: Applied Linear Algebra

Muhammad Nawaz Sharif University of Engineering & Technology

- Teaching Assistant, CSC-341: Introduction to Computing
- Teaching Assistant, EEE-562: Linear Control Systems

Islamabad, PK

Fall 2023

Multan, PK

Fall 2019

Fall 2020

SCHOLARSHIPS & AWARDS

- ACC 2026 Student Travel Grant.
- Studentship for the Model Predictive Control (MPC) Course by Prof. A. Bemporad, IMT School for Advanced Studies Lucca.
- Erasmus Mundus Scholarship for the Joint Master's Degree in Electric Vehicle Propulsion and Control.
- Awarded a laptop by the Prime Minister of Pakistan for maintaining a high GPA in Master's studies.
- Dean's Honour Roll – Fall 2019, Spring 2020, Fall 2020, and Spring 2021.
- HEC Merit-Based Scholarship.
- Scholarship recipient under the *SCHOLARSHIP-AMERICA* program by PepsiCo Foundation during B.Sc. Electrical Engineering.
- Best Final Year Project, Batch EE-2017.

PROJECTS *LINKS TO FULL REPORTS EMBEDDED

Model-Free Intelligent Control Design For Underactuated Mechanical Systems

-Master's Thesis

In recent years, there has been growing interest in the development of controller design and algorithms for underactuated systems. Underactuated systems are those in which the number of actuators is less than the number of degrees of freedom. This makes them more challenging to control, but they are also more common in practice. Model-free control techniques are attractive because they do not require any information about system's dynamics. In my research, I have developed a new control framework using model-free robust adaptive controller for underactuated systems.

Intelligent SOC-SOH Estimation for Li-ion Batteries

-E-PiCo+ Project

Designed and implemented LSTM models for SOC and SOH estimation with optimized computational efficiency for embedded BMS. Automated hyperparameter tuning via Optuna and conducted comparative analysis against FCN and CNN architectures to validate real-time performance.

Smart Bionic Prosthetic Leg (Best Final Year Project Award)

-Bachelor's Thesis

My Final Year Project is a Smart Prosthetic Leg which mainly focuses on patients of an above-the-knee amputee. In this project, we have designed a smart and safe embedded system that controls the leg using different modes for different activities. Due to the low cost of our design, this prosthetic technology could be applied in public sector hospitals to serve the less fortunate.

TECHNICAL SKILLS

Professional:	Communication Skills, Problem Solving, Data Management, Leadership
Programming:	Julia, Python, C/C++, Assembly Language, Ladder Logic Language(PLC), $\LaTeX$
Software & Tools:	MATLAB/Simulink, Git, ROS, Proteus, Multisim, LabVIEW, Microwind, Simcenter Amesim

LANGUAGES

• English

• French

• Urdu

REFERENCES

Dr. Usman Ali

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School of Electrical Engineering and Computer Sciences
National University of Sciences & Technology
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Dr. Rameez Hayat

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Dr. Muhammad Ali Murtaza

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Information Technology University (ITU), Pakistan
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